

	BOL 1022	Langeveld G224	SCG C028
	Stella Donker	Christoph Strauch	Jeroen Benjamins
12:20	Build up poster boards, Evenementen ruimte Langeveld		
13:05	Welcome!		
13:15-13:25	Slappendel, Pieter	Zhang, Bill	Chen, Celine
13:35-13:45	Verbeten, Femke	Sagou, Olga	Gunawan, Hendrik
13:45-13:55	Spekreijse, Pim	Sneep, Jeannelle	Beuparlant, James
Break until 14:10			
14:10-14:20	Mandias, Gregoria Frederika	Cabral Litowsky, Yanna	Jansen, Jolijn
14:20-14:30	Hajtić, Kasia	Hamadeh, Farah	Angeliki Konstioti
14:30-14:40	Kuggeleijn, Mats	Vandame, Lola	Wantenaar, Sophia
14:50	Sosnova, Angelika	Zon, Justin van	Lina Heidt
14:50-15:00	Qianzi, Lyu	Zofia Kotwica	Mc Meekin, Paul
Break until 15:10			
15:10-15:20	Aguirre Gomez, Camilo	Guarisco, Manfredi	Dzięgielewski, Stanisław
15:20-15:30	Klaren, Carmen van	van der Loo	Tjio, Surya
15:30-15:40	Karioti, Kallia	Huang, Po-Hsuan	Marel, Tanja van der
15:40-15:50	Malvido Monta, Zoe	Groenewoud, Florence	Baarda, Ydo
15:50-16:00			Angie Mehta
16:00-16:10			
16:00	Walk to Langeveld/poster session		
16:00-17:30	Poster session		
17:30-	Drinks, snacks, and cookies		
Video presentations by	Pedro Trueba Plaza Tiffany Chen Xia Lingtao – sends in later Daan Amelink		

posters by

Streefkerk, Vivianne
Slors, Tycho
Silberberg, Sasha
Riba Ustrell, Ivet
Zhang, Xi (Naomi)
Teinsma, Esmée
Graaf, Janneke van de
Thorwart, Raphael

attend presentations in

Bol 1.022
Bol 1.022
Bol 1.022
Bol 1.022
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Bol 1.022
Bol 1.022

Baars, Mirte	Bol 1.022
Chen, Tiffany	Bol 1.022
Heuvel, Anneliek	Bol 1.022
Andrikopoulou, Alisso	Bol 1.022
Weering, van (recidivist)	Bol 1.022
Langenhuijsen, Elsemieke	Langeveld G224
Turjaka, Vesa	Langeveld G224
Thalita, Nabila	Langeveld G224
Min, Ebelien	Langeveld G224
Muus, Ottavio	Langeveld G224
Olympios, Michail	Langeveld G224
Kotwica, Zofia	Langeveld G224
Jose Trueba Plaza	Langeveld G224
Schout, Julia	Langeveld G224
Zijtveld, Emmie van	Langeveld G224
Nanninga, Sophie	Langeveld G224
Kondratyev, Nikita	Langeveld G224
Gu (Robin), Zirui	Langeveld G224
Amjad	SCG C028
Mehta, Angie	<i>Presents too</i>
Coleman, Shayna	SCG C028
Kontsioti, Angeliki	<i>Presents too</i>
Brust, Loes	SCG C028
Veldhuis, Shawn	SCG C028
Safar, Zainab	SCG C028
Cotelo García, Daniel	SCG C028
Lopez Rodriguez - Carretero, Fernando	SCG C028
Karioti, Kallia	SCG C028
Dzięgielewski, Stanisław	SCG C028
Boogaard, Juul van den	
Ditmarsch, Mark van	Langeveld
Balali, Sofia	SCG
Zhang, Shiyi	

Name Student: Tiffany Y. Chen

Internship or Thesis: Thesis

Title project: Eye tracking: can intrinsic values of the items affect people's gaze trajectory?

(Abstract, 100-150 words)

The human gaze is a powerful tool for understanding how people perceive and interact with each other and the world around them, which has huge potential yet to be discovered. The gaze trajectory, or the path that the eyes move from one point to another, is particularly informative. We have known that when executing an eye movement to a target location, the presence of an irrelevant distracting stimulus can influence the saccade metrics and latency. In this thesis, we are trying to investigate if distractors' differences in intrinsic values would affect the saccade trajectory. In the experiment, the target appears in horizontal line from left or right, and the distractor's location can be either remote or close.

My hypothesis is that the monetary intrinsic value of the distractors would affect the saccade trajectory. 12 stimuli B/W pictures were selected to be the distractors of the experiment as they meet the criteria of clear distinguish of low and high intrinsic value. Questionnaires were conducted to verify the choices. All the stimuli pictures were processed as scrambled random pixels, keeping the same number of pixels and size of the pictures as the stimuli. The possible targets and distractors' locations are shown as below illustration. The experiment consists of 6 blocks including a practice block. The participant is instructed to fixate their eye in the centre fixation point until the dot becomes black (when the target shows up), then making the eye movement as soon as possible. In total, each participant finished 1200 trials (including the baseline condition where no distractor will appear). In the data analysis process, I investigate both saccade start and end point of each trial with associated target and distractor location; center of gravity points for saccade trajectory; baseline condition and repeated measure ANOVA conducted for each participant.

Name company or University: Utrecht University

(If applicable) supervisor:

UU Supervisor: Jessica Heeman

Name Student: Hendrik Gunawan
Thesis: Oral Numerosity Perception

(Abstract, 100-150 words)

Numerosity perception is a basic human ability to represent a quantity without counting. In numerosity perception, there are three ways to enumerate objects: subitizing, counting, and estimation. Previous research had investigated the capability of numerosity perception thorough visual, auditory, and haptic means. However, it had not delved into the ability of numerosity perception thorough oral. Even though oral perception is part of haptic perception, there was no previous research investigating numerosity perception particularly subitizing by oral means. The closest previous research to oral numerosity perception was subitizing on active and passive touch thorough hands. Current study collected 25 participants in order to see the numerosity perception in mouth. The objects used to enumerate were Hands off My Bite Crispy. Average volume and weight of these candies were respectively 39.5 ml and 1.364 gram. Participants were given a paper cup which contained 1-9 candies randomly. The data result showed that numerosity perception was possible in mouth. In addition to that, the limit of oral subitizing was 3. It was similar to numerosity perception thorough hands.

University: Utrecht University
UU Supervisor: Dr. Garnt Dijksterhuis

James Beauparlant

Thesis Abstract

The present study aimed to investigate the effects of exercise as a lifestyle factor on cognitive control and impulse control in healthy adults. Primary research questions focused on the relationship between exercise, cognitive control, stopping success as measured by EEG during a stop-signal task, and impulse-control failures. Secondary research questions examined the differences in EEG alpha oscillations over the visual and auditory cortex between people who exercise and those who do not, specifically assessing whether the topographical information provided by the stop task indicates performance differences mediated by exercise interventions. Tertiary research questions addressed how exercise affects inhibitory control, as measured by theta oscillations in the prefrontal cortex during a stop task. Participants were recruited in person, through word of mouth, and through local advertisements. Participants completed a stop-signal task while their EEG activity was recorded. Results showed that individuals exhibited differences in alpha oscillations over the visual and auditory cortex. Finally, theta oscillation changes in the prefrontal cortex during the stop task were also observed. These findings suggest Theta oscillations may act as a predictive marker for cognitive and impulse control in healthy adults.

Sasha Silberberg

Thesis

**The Comparative Effects of Silence, Birdsong, Music, and a Medical Soundscape on
Cognitive Performance and Perceived Workload**

Medical soundscapes (e.g., rattling trolleys, alarms) may degrade cognitive performance in a hospital setting. Here, I assess the impact of medical, birdsong, and music soundscapes on the recall of numerals projected on a monitor for 210 ms before being masked. Participants completed the memory task while being exposed to one of three soundscapes: birdsong, a Mozart sonata, or a medical soundscape. A 2x3 design was used, where sound condition (sound versus silence) was the within-participant manipulation and the between-group variable was the soundscape type. Participants also completed a questionnaire focused on perceived workload, the pleasantness of the soundscape, individual sensory-processing sensitivity, and other background information. A repeated measures analysis of variance showed no statistically significant difference in performance between all three groups of soundscapes. Despite the medical soundscape's rating as the most unpleasant soundscape, its perceived workload scores were the lowest. This finding may be attributable to the type of task or auditory stimuli used.

TU Delft Faculty of Industrial Design Engineering

Dr. Elif Özcan Vieira

Dr. Martijn Mulder

Name Student: Fernando López Rodríguez-Carretero

Internship or Thesis: Internship Human Factors Research (12.5 ECTS)

Title project: Distinguishing HMI for ADAS and ADS

The aim of the internship assignment is focused on assisting RDW to finalise the projects on *Distinguishing HMI for ADAS and ADS* and the *HMI Benchmark* within the Department of Applied Innovation (Human Factors). More specifically, it will include revising the *quick response* task of the experiment setting as one of the parts of the whole study. It will entail turning changes into an online study, running the study, analysing the data and drafting it into a report. In parallel, I will be part of the meetings with the individual OEMs to discuss current findings and how they compare to their own HMI.

Name company or University: The Netherlands Vehicle Authority (RDW)

(If applicable) External supervisor: Ilse Harms

UU Supervisor: Stella Donker

Name Student: Camilo Aguirre Gómez

Internship or Thesis: Internship

Title project: Testing the most effective communication to increase engagement in feedback.

This internship focuses on conducting a user experience (UX) research project to determine the most effective way to engage users as participants in providing feedback for a platform that develops low-code tools. The study aims to compare two different messaging approaches: a pop-up within the platform and an email campaign. By testing these two methods, the internship seeks to understand which kind of invitation yields higher engagement rates.

The significance of this project lies in the necessity to expand the panel of participants for various UX research projects. Increasing the number of participants will enhance the diversity and representativeness of feedback received, leading to more robust and insightful findings. The findings from this internship will provide valuable insights into the most efficient and effective strategies for inviting users to participate in research activities, ultimately contributing to the growth and success of the platform's UX research initiatives.

Name company or University: Mendix

(If applicable) External supervisor: Ana Steblaj

UU Supervisor: Sjoerd Stuit

Name Student: Qianzi Lyu

Internship or Thesis: Thesis

Title project: Can intrinsic value influence the saccade endpoint?

Abstract: The visual world we encounter in everyday life is complex and filled with a large amount of information. Saccadic eye movements are made to shift the gaze to an object in a visual scene. When a saccade is executed, its endpoint, therefore, reflects the outcome of a selection process of all possible target locations. In our present study, we investigate how the intrinsic value and distance of distractors influence the saccade endpoint. We use objects with different monetary value to manipulate the saccade. The result showed saccade latency is different in conditions of high value and low value distractors. We are still processing the data of saccade endpoint.

Name Student: Po-Hsuan, Huang

Internship or Thesis: Thesis

Title project: An empirical investigation of individuals' recognition performance in relation to posterized and non-posterized images.

In this presentation, we will focus on a picture recognition experiment which aims to examine the difference between people recognizing two types of images: posterize and non-posterize.

In this experiment, we applied three hundred and twenty photos. All the images have a variant which is posterized. Participants first viewed the target images and then selected the presented images in the test phase based on a two-alternative-forced-choice structure.

We use posterization tools in photo editing software because it is a reasonably objective method for manipulating the degree of colour separation compared with hand drawings.

We will mainly introduce the details of the experiment design and some related information. As we are still analyzing data, some analysis techniques may be discussed.

Name company or University: Utrecht University

(If applicable) External supervisor: N/A

UU Supervisor: Suzanne Hut

Thesis abstract for Update Day

Vesa Turjaka (2335174)

Thesis (27.5 ECTS)

Title: What happens to trust when a robot partner does something unexpected: the role of transparency and outcome

The development of trust is crucial for effective performance and collaboration in Human-Robot teams. Communication plays a vital role in establishing trust, particularly when the robot exhibits unexpected behavior that could degrade trust. Transparent communication enables the human to understand the rationale behind robot's decisions, preventing a significant decline in trust. This study investigates explanations for deviant behavior exhibited by robots when delegated tasks. Data were collected in a military mission-like environment, involving 82 participants collaborating with a robot teammate. Trust development was measured at various timepoints. The results demonstrate a robust effect of transparent communication on trust development. The robot's explanations, indicating high transparency, were particularly effective in mitigating trust decline. Conversely, trust diminished when the robot behaved unexpectedly without providing an explanation. These findings emphasize the importance of transparency in fostering trust within Human-Robot teams. Further research is necessary to generalize these findings to diverse contexts and populations.

Keywords: Human-robot team · Autonomous system · Trust · Task delegation · Transparency

Name of company: TNO

External supervisor: Prof. Dr. José Kerstholt

UU Supervisor: Dr. Christian P. Janssen

Name Student: Sophie Nanninga

Thesis MSc Applied Cognitive Psychology

Title project: **Immersive insights: a comparison of a 360-degree video and images with focus on emotional transitions after visualising plastic pollution.**

Abstract

This study investigates the efficacy of emotional interventions, specifically an immersive video and images about plastic pollution, in evoking emotional responses. Drawing on the concepts of environmental values and emotions, we examine how these interventions can influence pro-environmental behavioural change with a focus on one's biospheric values. A significant difference between the immersive video compared to images was expected, with a moderating effect of individuals' biospheric values. We found a significant different emotional change in the immersive video compared to images for seven emotions. The biospheric values were more impactful for most emotional changes compared to type of intervention, meaning that people are not likely to be affected by any intervention if they do not prioritise the environment. This research contributes to the broader field of environmental psychology and offers guidance for future investigations on the effectiveness of emotional interventions in environmental contexts.

Universiteit Utrecht

UU Supervisor: Garmt Dijksterhuis

Name Student: Xiyin Celine Chen

Internship or Thesis: Thesis

Title project: What does your nose really know? The effects of social norms and dietarian identity on odor perception.

Interest in vegetarianism and veganism as food choices and social identities is increasing but widespread adoption remains limited. To promote plant-based diets, novel strategies are needed. This study explored the influence of social norms and dietarian identities on olfactory perception. Data from 64 participants were collected using a mixed design. Participants rated odors labeled as „favored by vegetarians/vegans“ or „by omnivores“, compared to no labeling, and completed the Dietarian Identity Questionnaire (DIQ). Findings revealed that meat consumers rate meat and dairy odors higher than meat reducers, and the strength of dietarian identity influenced the impact of labels. Specifically, the omnivore-favored labeling on meat odors corresponded to a higher hedonic rating for meat consumers. However, no other expected label effects were observed. These results offer new impulses for future research on leveraging normative labels and dietarian identity to predict and influence odor perception.

Keywords: social norms, dietarian identity, food choices, odor perception

Name company or University: Utrecht University

UU Supervisor: dr. Garmt Dijksterhuis



**Utrecht
University**



**VRIJE
UNIVERSITEIT
AMSTERDAM**

Faculty of Social and Behavioral Sciences

Department of Education and Psychology

The Relationship between depression and social support among youth refugees

Zainab Safar (2317435)

Internal Supervisor: Larissa Meijer
Alözkan

l.l.meijer@uu.nl

external Supervisor: Cansu Sever

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Applied Cognitive Psychology

Master's Thesis [27,5 ECTS]

Utrecht University

Academic Year 2022/2023

Abstract

Over the past decades, there has been a steady increase in the number of people forced to leave their homes and seek refuge worldwide. The reasons for their displacement vary, but regardless of reaching a safe country, they continue to face significant daily challenges both mentally and physically. Additionally, a substantial portion of these refugees are young individuals who are at a high risk of developing mental health disorders due to their past hardships and the lack of social support. This study aims to gain a better understanding of the relationship between depression and social support among young refugees between the ages of 16 and 25, who speak Arabic, Ukrainian, or Tigrinya, and are currently residing in the Netherlands. The data for this study comes from a cross-sectional survey named as ReYoUnite which uses a user-friendly digital questionnaire. Participants recruited via several methods, such as social media, stakeholders working with refugees, and schools. The target group is refugee youth between 16-25 years old who speak Arabic, Ukrainian and Tigrinya languages and live in the Netherlands. The participants give permission via the digital tool and fill in the survey. After that, they answer self-report questionnaires, the Kessler Psychological Distress Scale (K10), WHO Disability Assessment Schedule 2.0 (WHODAS 2.0), Hopkins Symptom Checklist 25 (HSCL-25), PTSD Checklist DSM-5 (PCL-5), Post Migration Living Difficulties Checklist (PMLD), Trauma Experiences Checklist, and Multidimensional Scale of Perceived Social Support (MSPSS). Questionnaires will be available in the target group's mother language. Taking part in this survey is free, and they can stop at any time without clarifying or skipping a particular question. The survey will take approximately 30 minutes to fill in. The pseudonymised data will be analyzed using SPSS Software. This study is one of the fewer studies focusing on the relationship between social support and depression for young refugees in the Netherlands and we believe it is important to increase awareness for youth refugees' mental health and how to provide better care.

Kallia Karioti (4431391)

Thesis project

Mixed Reality in Cognitive Rehabilitation

(Abstract, 100-150 words)

Utrecht University

UU Supervisor: Charlotte van de Wouw

The title of this thesis project is “Mixed Reality in cognitive rehabilitation”. Its main goal is to find out the importance of Mixed Reality (MR) in cognitive rehabilitation, specifically cognitive assessment, focusing on the Greek healthcare system. Moreover, in order to find out what exactly MR provides to therapy, and its potential barriers for implementing MR in Greek workplaces is necessary to gain insight into the potential different perceptions of healthcare professionals (e.g., clinicians and researchers) towards MR in cognitive assessment and whether they have an experience using MR during their career. Taking all the above into consideration, many different definitions and perceptions were observed. Lastly, through this thesis project Greek and Dutch data are going to be compared in order to find out potential differences between these two healthcare systems. The main research questions are the following: “Which are the potential barriers in order to implement MR in cognitive assessment?”, “What changes should happen in the healthcare system in order to implement MR in cognitive assessment?” and “What are the potential differences between Greek and Dutch healthcare system towards MR?”. An online survey is the research tool which was used for this project. This survey consists of 20 questions, about the potential experience that Greek healthcare professionals have with Mixed Reality, how useful or not they think Mixed Reality is in cognitive assessment and why they are or not using Mixed Reality in their workplaces. Concerning the Greek data there were 49 respondents, all of them Greek healthcare professionals and their answers were recorded through Qualtrics. Last but not least, the method that was used for analyzing the recorder data was qualitative method.

Name Student: Gu Zirui

Internship or Thesis: Internal Combined Thesis

Title project: Investigating Pseudoneglect across Varies Distance with Pupillometry in Virtual Reality

Pseudoneglect has been investigated under many disciplines. Research has shown that distances can act as a modulation toward pseudoneglect (Heber et al., 2010). However, it is difficult to design paper and pencil tasks in various distances. Virtual reality technique provides us an opportunity to create a three-dimensional environment to present stimuli in different distances. In our study, we developed a pupillometric test in VR environment to investigate pseudoneglect in four distances (30cm, 70cm, 120cm, 700cm). 34 healthy participants are instructed to complete a VR eye tracking experiment where stimuli in eight conditions were randomly presented. The result demonstrates a slight but not significant left bias in healthy control, which is not sufficient to reflect pseudoneglect. We also found that the leftward bias tended to decrease as viewing distance increased, but the differences are not significant, too. Future studies are required to replicate our experience using VR eye tracking technique.

Name company or University: Utrecht University

UU Supervisor: Dr. Christoph Strauch

Name Student: Shiyi Zhang

Internship: Customer Experience Optimization at

Title project: Enhancing User Satisfaction: Revamping Customer Experience in an E-Business Environment

As a Customer Experience Optimization Specialist at Heineken, my role involved enhancing customer interactions and increasing conversion rates in the e-business domain. Key aspects of my work included A/B testing, usability testing, and conversion-focused strategies. Through A/B testing, we compared different web page versions and user experiences to identify optimal configurations for higher conversion rates. Usability testing helped uncover pain points and improve the user experience. By optimizing website design, simplifying interfaces, and personalizing interactions, we aimed to create a seamless customer journey. Collaboration with cross-functional teams allowed for data-driven decision-making and the implementation of customer-centric solutions. Continuous monitoring and analysis enabled us to refine strategies and drive business growth. Ultimately, my role focused on optimizing the e-business ecosystem, leveraging customer insights, and improving conversion rates for enhanced customer satisfaction.

Name company: Heineken N.V.

External supervisor: Adriana Ivascu

UU Supervisor: Garmt Dijksterhuis

Name Student: Olga Sagou

Internship or Thesis: Thesis

Title project: What's in the box? The Role of Auditory and Haptic Cues on Object Enumeration within Containers

This study investigated the influence of auditory and haptic cues on quantity estimation of items (beads) inside handheld boxes. Three conditions were tested: auditory-only, haptic-only, and combined auditory-haptic cues. Results showed that the auditory-only condition yielded the most accurate estimations, followed by the haptic-auditory condition, and the haptic-only condition. The termination point of subitizing was around 3 objects for auditory-only and haptic-auditory conditions, while the haptic-only condition terminated at 2 objects. Auditory cues played a dominant role in estimations, and the integration of sensory cues improved performance compared to relying solely on haptic cues. Participants exhibited an underestimation bias for larger numbers of beads across all conditions. The haptic-only condition had higher error rates and lower accuracy compared to the auditory-only condition, supporting the limitations of haptic cues. These findings contribute to understanding multisensory integration and the impact of sensory cues on numerical estimation.

Keywords: multisensory integration, auditory cues, haptic cues, estimation, subitizing, numerosity.

Name company or University: Utrecht University

UU Supervisor: Krista Overvliet

Name Student: Lingtao Xia

Internship or Thesis: Internship

Title project:research assistant at Radboud University

I will work with three Phds to help them with their study and learn some precious experience through this process. My work will including but not limited to: data analysis, coding, writing reports and conducting experiment for data collection.

Name company or University:Radboud University

(If applicable) External supervisor:Dr.Barbara Muller

UU Supervisor:Klink, P.C.(Chris)

Poster

Name Student: Zofia Kotwica
Internship or Thesis: Thesis (27.5 ECTS)

Title project: The mediating effect of response inhibition on the relationship between dieting preoccupation and impulsivity.

(Abstract, 100-150 words)

With thinness often viewed as the only desirable body type, the dieting preoccupation is becoming more prominent. Concurrently, obesity levels are growing continuously. Many articles suggest that excessive thinking about dieting is ineffective or counterproductive in losing weight. Scientists indicate impulsiveness and disinhibition as possible causes for this phenomenon. To explore the linkage and find potential tools for determining the risk of unhealthy food preoccupation, we investigated if inhibitory control abilities mediate the relationship between food preoccupation and impulsivity. The Stop-Signal Task (SST) was used as a measure of inhibitory control, the Revised Restraint Scale (RRS) and Food Frequency Questionnaire (FFQ) measured dieting preoccupation, and Barratt Impulsiveness Scale 11 (BIS) measured impulsiveness levels among the adult participants. The regression analysis did not demonstrate significant correlations between any of the measures except for RRS and FFQ, and therefore the mediation of inhibition on impulsiveness and dieting preoccupation was not found.

Name company or University: Utrecht University
UU Supervisor: Daan van Roij

(They are very similar due to the same methods, and the same day of presenting)

Presentation

Name Student: Zofia Kotwica
Internship or Thesis: Thesis (27.5 ECTS)

Title project: The mediating effect of response inhibition on the relationship between dieting preoccupation and impulsivity.

(Abstract, 100-150 words)

Scientists suggest that impulsiveness and disinhibition might be possible causes for constant dieting not being linked to weight loss, increasing impulsive eating, and making sustaining a desirable diet difficult. To explore this dependency and find potential tools for determining the risk of unhealthy food preoccupation, we investigated if inhibitory control abilities mediate the relationship between food preoccupation and impulsivity. The Stop-Signal Task (SST) was used as a measure of inhibitory control, the Revised Restraint Scale (RRS) and Food Frequency Questionnaire (FFQ) measured dieting preoccupation, and Barratt Impulsiveness Scale 11 (BIS) measured impulsiveness levels among the adult participants. The regression

analysis did not demonstrate significant correlations between any of the measures except for RRS and FFQ, and therefore the mediation of inhibition on impulsiveness and dieting preoccupation was not found. We suggest that low levels of food preoccupation might be necessary to sustain healthy eating behaviors.

Name company or University: Utrecht University
UU Supervisor: Daan van Roijj

Name Student: Pedro José Trueba

Internship or Thesis: Thesis

Title project: Comparative Analysis of Behavioral Elicitation Methods in Understanding Decision-Making Processes

The objective of this research is to understand the process of decision making in an economic task. Researchers have observed that a single behavioral elicitation method does not provide sufficient insight into the cognitive processes involved in decision making. In the case of the self-report methodology, where participants explicitly report on the strategies they employ, it has been argued that participants often struggle to access or accurately report on their strategy generation. On the other hand, alternative methods, such as the Hidden Markov model, have been developed to identify hidden strategies based on observable behaviors like reaction time and accuracy. However, this method assumes that each set of behaviors corresponds to a specific strategy, which may not always be the case. Considering the disadvantages of both methods while recognizing their validity to some extent, the goal is to compare them through an experimental design.

Name company or University: Utrecht University

UU Supervisor: Leendert van Maanen

Name Student: Xi Zhang (5456932)

Internship or Thesis: thesis

Title project: Desires for high-calorie foods in imaginative and immersive virtual reality winter environment contexts

Anything that allows subjects to mentally 'leave' the real-world setting and enter the realm of their imagination can be seen as an 'immersive' technique. Specialized surveys have been developed to detect the consumers' level of 'immersion' in such situations (Dijksterhuis et al., 2022). Virtual Reality (VR) using head-mounted displays now can closely mimic real-world conditions, offering unprecedented opportunities to understand human cognition and behavior beyond any research environment (Hardiess et al., 2015). Immersing users in realistic yet controlled simulation scenarios create an authentic testbed for proactive behavior interventions and support. VR has become an essential tool for studying food consumer behavior in the psychology field because of these benefits. In contrast to VR as a high-end technology, having participants look at a picture or listen to a good story in a quiet environment that fully engages the imagination of the test subject can also be seen as an immersive 'technique' by imagination and low-tech.

Existing research has demonstrated that watching videos depicting winter cues elicits thoughts about energy-dense foods and survival (Folwarczny et al., 2022). Such cues elicit a higher preference for energy-dense foods over low-calorie foods, and this effect may differ between women and men. Another study (I. G. Andersen et al., 2018) has demonstrated a significantly stronger desire for cold beverages compared to hot ones during beach exposure in VR and photo-enhanced imaginative (PIC) conditions. Further, desires for cold drinks were significantly greater in the VR simulation compared to the PIC condition. These results suggest that contextual exposure can induce contextually congruent desires and likeness and that VR has the potential to evoke greater engagement and affective responses compared to traditional media.

This study aims to explore the extent to which VR can elicit desires for high-calorie foods compared to more traditional techniques (more specifically, pictorial representation combined with storytelling) in winter environments. The results will demonstrate whether contextual exposure through immersive virtual reality (VR) or photo-story-enhanced imaginative conditions can induce contextually congruent desires and preferences.

Name Company or University: Utrecht University

(If applicable) External supervisor: \

UU Supervisor: Garnt Dijksterhuis

Name Student: Ottavio Muus

Internship or Thesis: Thesis

Title project: The working memory capacity of the tactile sense in relation to the usage of tactile maps for blind people.

Abstract:

This study investigates how many temporary bindings between tactile symbols and nouns of a length of one syllable can be bound and correctly recalled immediately after. The importance of this lies in the urge to increase the usability of tactile maps for blind or visually impaired people, since legends of tactile maps contain a bunch of tactile symbols which have to be learned and recalled during the usage of the tactile map. Knowing more about the capacity limit provides possible insight for increased applicability of tactile maps. During the task blindfolded participants had to carry out, arrays of tactile symbols had to be felt, while simple words were spoken. These were immediately tested in a recall test with the same set of symbols, but in a different order. These arrays varied chronologically from a length of 3 symbols to a length of 10 symbols. For arrays of all length, 3 symbols were on average correctly recalled.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Krista Overvliet

Name Student: Daan Amelink
Internship or Thesis: Internship
Title project: Eye tracking

Eye tracking is a relatively modern technique which is utilized to gain insight into individuals' gaze behavior. Besides estimating gaze position, eye tracking can be employed to collect a vast amount of eye metrics that hold information about a person's cognitive state. For instance, it has been reported many times over that pupil size positively correlates with mental workload. To test our cost-effective, less expensive Pupil Core eye tracker can consistently measure subtle differences in pupil size as a result of a change in mental workload, we build an digit span computer experiment. During the process, we ran into numerous challenges that gave we to new learning experiences. All the lessons learned while setting up and running this pilot experiment will be condensed into an advisory report.

Name company or University: Thales
(If applicable) External supervisor: Iris Cohen
UU Supervisor: Leon Kenemans

Name Student: Jolijn Jansen

Internship or Thesis: Thesis

Title project: Point symbol discrimination on a tactile map

In the Netherlands 300.000 people have a visual impairment, which can debilitate their spatial orientation and mobility. To facilitate orientation a large scale project was started to create accessible, automatically-generate tactile maps of Dutch neighbourhoods. Part of this project is to investigate which tactile point symbols are best to distinguish and differentiate and which point symbol features could increase differentiability. Fifteen participants were asked to perform a tactile discrimination test, describe symbol outlines and answer questions about their experience with a prototype map. It was found that 5 out of 9 symbols were distinguishable and 7 out of 9 symbols were differentiable. From the symbol features form, fill, size, and interruption, none could independently improve symbol differentiation. The experiment yielded a symbol set of eight point symbols expected to be differentiable on a tactile map. Yet, future research is needed to confirm usability and user-friendliness of this improved symbol set.

Name company or University: Stichting Accessibility

External supervisor: Aafke van Welbergen MSc

UU Supervisor: Dr. Krista Overvliet

Name Student: Mark van Ditmarsch

Internship or Thesis: Internship

Title project: Determining the range of the Emotion Impact Score

Bloakes has developed an Emotion Impact Model that employs association sets and implicit research methods to quantify the customer journey in terms of emotions. A higher Emotion Impact Score (EIS) indicates stronger associations and a greater likelihood of desired customer behavior. Currently, Bloakes is capable of solely providing clients with a relative EIS report, which involves tracking scores over time and analyzing the customer journey. However, Bloakes aims to offer clients comprehensive insights regarding an absolute EIS. To achieve this objective, this research sought to establish a practical range through a binary choice task involving two positively perceived brands (Nike and Coca-Cola) and two negatively perceived brands (Shell and TATA Steel). The results of the study aligned with expectations, with Nike and Coca-Cola exhibiting high EIS and Shell and TATA Steel displaying low EIS. Although the scores could have been slightly higher with a more specialized association set, an approximate upper bound based on Nike's score would be 60, while an approximate lower bound based on TATA Steel's score would be -20.

Name company: Bloakes

External supervisor: Rianne Besters

UU Supervisor: Daan van Rooij

Name Student: Sophia Wantenaar

Internship or Thesis: Internship

Title project: The Public Perception of Hydrogen

Hydrogen is expected to play a prominent role in the realisation of a carbon-neutral energy system. HyScaling, a research consortium consisting of several partners, aims to establish a Dutch electrolyser industry and green hydrogen economy. Electrolysers split water into hydrogen and oxygen, using electricity. When using sustainable energy resources, e.g. from solar and wind, no CO₂ is emitted. This technology is not yet available on larger scale. However, the demand is expected to increase enormously, given the climate goals for 2030 and 2050. To secure the Netherlands with a strong position in the hydrogen market, scaling up electrolyser and hydrogen production is needed. Insights on the public perception of the Dutch population regarding green hydrogen can aid investors and policy makers with decision making on related choices still on the table. Therefore, a questionnaire was designed regarding this topic during the internship. Therein respondents are presented with information about those choices, and are asked about their views on them.

Name company or University: TNO

(If applicable) External supervisor: Melanie Klösters

UU Supervisor: Sjoerd Stuit

Name Student: Elsemieke Langenhuijsen

Internship or Thesis: Thesis

Title project: Unlocking job market potential: A website usability study

Abstract:

The job market in the Netherlands is historically tight and employers have to compete for talent, but candidates need to be attracted and motivated to apply. Previous research has found evidence for the importance of usability in the process of online recruitment. Studies found that usability can help enhance the organizational image and recruitment effectiveness and that website usability can be optimized using design principles. During this research, three studies have been conducted to examine interface design modifications. The first two studies explored the potential usability issues on the website and have shown that participants encountered issues regarding the ease of understanding and the ease of navigation of the website. Four prototypes of the website with different modifications were created and tested on constructs of usability. The study showed no significant differences among the different prototypes on all constructs of usability but still yielded some interesting findings.

Name company or University: Direct Research

(If applicable) External supervisor: Carlijn van de Pas

UU Supervisor: Christoph Strauch

Name Student: Vivianne Streefkerk

Internship or Thesis: Thesis

Title project: Twice as Nice with Half the Effort: The Developer's Guide to Reusable UX/UI Design Systems in Mendix

A reusable design system provides guidance for developers to create better applications in terms of UX and UI. A design system improves consistency within and across applications, decreases the time spent developing and makes implementing UX and UI more accessible for developers with limited knowledge on these topics. To investigate the necessary components of a reusable design system, interviews were carried out to answer the question: 'What are the fundamental aspects to consider from a developer's perspective when creating a reusable UX/UI design system?'. According to developers, the fundamental aspects of a reusable UX/UI design system should be applied in a two-fold setup, with a style guide on the design of elements and substantiated guidelines on how to implement the elements. Combined with theories from cognitive psychology on UX and UI design, a proof of concept for the design system is created.

Name company: Capgemini Nederland

External supervisor: Youp Macrander & Matthijs Lok

UU Supervisor: Dr. Robert Weijers

Samuel van der Loo
Internship

Sound Implementation on the Phone: Promoting Customer Patience during Waiting Time in the Customer Service of Banks

During my internship at ABN AMRO Bank N.V., I had the opportunity to work on the UX Design of customer interactions, specifically focusing on telephone-based services. One prevalent issue faced by customers is the long waiting times for customer service, leading to frustration and potential abandonment of calls. Recognizing the importance of addressing this challenge, my project centered around enhancing the wait experience, particularly using wait music. The aim of my project was to investigate the impact of different sound implementations on customers' patience while waiting for phone-based customer service. Drawing upon existing literature on enhancing customer patience, a survey and interviews were conducted to assess various sound implementations and capture the customer's waiting situation. The findings revealed that music, particularly cheerful music, had the most positive effect on participants' patience levels. Furthermore, it was observed that customers predominantly wait in private settings with their phones on speaker.

ABN AMRO Bank N.V.
Femke van Drooge
Sam van Bijnen

Name Student: Ebelien Min

Internship or Thesis: Thesis

Title project: Braille proficiency in tactile map users

Expert organisation *Accessibility* is aiming to create a prototype tactile map on a neighbourhood level that is suitable for blind braille users. In later stages of the development, this prototype will be adapted so it can also be used by other visually impaired people (VIP). To acquire more insights on the useability when designing a tactile map, this study looked at the differences the acquisition age of braille has (if any) on the understanding and useability of a prototype tactile map. The participants were split into two groups, one with participants who learned braille at a young age (3-8 years old), and one with participants who learned braille at a later age (14+ years old). The participants performed different search tasks on a prototype map, during which their search times were measured, as well as their error rates. Group 1, who learned braille at a young age, showed a faster response time (average of 530 ms per question) than group 2, who learned braille at a later age (average of 1280 ms per question). Group 1 also showed a lower error rate (average of 4.7 out of 12 questions wrong) than group 2 (average of 5.8 out of 12 questions wrong). Both the error rates and the response times did not differ significantly from one another in group 1 and 2, but it does emphasize the diversity of the group of VIP once again. This diversity shows that some form of additional research is necessary to provide more insights on the differences in needs among the users of a tactile map. This way the tactile map will hopefully fill in the final piece of the puzzle to help VIP with navigating from point A to point B.

Name company or University: Stichting Accessibility

(If applicable) External supervisor: Aafke van Welbergen

UU Supervisor: Krista Overvliet

Name Student: Femke Verbeten

Internship or Thesis: Thesis

Title project: A Different Perspective: The Role of Traffic Sign Orientation in Message Transfer at Parallel Lanes

In this study, it was experimentally evaluated how different lane positions and sign orientations affect traffic sign message transfer at parallel highway lanes. The reason for the study was that qualitative studies gave indications that road users experience uncertainty when choosing a driving speed in certain parallel lane traffic situations. One thing that might reduce doubt is rotating a traffic sign horizontally towards the lane for which the message is intended. Whether the traffic sign orientation or the lane position affects the message transfer was not known. Therefore, this was investigated using an online reaction time experiment in which participants chose whether the traffic sign message was intended for their lane or not after seeing traffic stimuli. This study can help road and signage designers create safe designs.

Name Company or University: Rijkswaterstaat

External supervisor: Anke Mulder & Chantal Merkx

UU Supervisor: Chris Janssen

Name Student: Esmée Teinsma

Internship or Thesis: Thesis

Title project: Motivation to eat meat and animalic odour preference: Exploring the influence of animal product consumption on animalic odour rating

(Abstract, 100-150 words)

The degree to which individuals are motivated to eat meat and other animal products may influence how odours of animal origin are perceived. This study examines the relationship between individuals' motivation to eat animal products and their perception of animalic odours. The Motivations to Eat Meat Inventory is used to measure the willingness of individuals to eat meat and other animal products. Participants provided ratings based on the pleasantness, edibility, willingness to eat, intensity, and familiarity, for both animal product odours and non-animal product odours. Significant effects were found for pleasantness and willingness to eat for meat and fish odours, with higher MEMI scores associated with increased ratings. However, for dairy and egg odours, no such effect was found. These findings suggest that the motivation to eat animal products is related to the preference for meat odours, but not necessarily for all animal products.

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Dr. Garmt Dijksterhuis

Name Student: Anneliek Heuvel

Internship or Thesis: Internship 12.5 EC

Title project: Behaviour and Biodiversity: Factors that Influence Engagement of Pro-biodiversity Behaviours in urban Environments

Abstract (100-150 words): The planet is facing a worldwide biodiversity crisis. In urban environments residential green areas have an important role to play in supporting biodiversity. It is important to gain insight into what factors influence behaviours that support biodiversity in urban environments. This literature review provides insight into factors that influence individuals' behaviours that support biodiversity in urban environments. Scientific databases were consulted and yielded 17 articles. The identified factors were integrated into the COM-B model of behaviour. The aim of this review is to provide insight into which factors are important to consider when engaging citizens in behaviours to support biodiversity within urban environments.

Name University: Hogeschool van Amsterdam

External supervisor: Dr. Loes Kremmers

UU Supervisor: Charlotte van de Wauw

Name Student: Pieter Slappendel

Thesis: EEG and Inhibitory Control

Title project: Predicting cognitive failures in impulse control

Cognitive control is important in the daily life of people. Especially the ability to dynamically adjust one's behavior in everyday activities, such as driving and sports. This is called inhibitory control; one's ability to stop an action that is already in progress. Proactive response inhibition happens when one prepares to withhold a reaction without the stimulus having appeared yet. The current research serves as a further examination of proactive response inhibition by measuring theta band activity in the brain with EEG while participants do a stop-signal task. The findings suggest that there is a significant effect of theta on task performance. Yet, there were no further significant results found.

Name company or University: Utrecht University

UU Supervisor: Sam van Bijnen

Name Student: Janneke van de Graaf

Internship or Thesis: Thesis

Title project: The effect of cognitive load on the effectiveness of a type 1 and a type 2 nudge and its consequences for the perceived autonomy

Earlier research has shown that type 2 nudges make use of reflective thinking, where type 1 nudges do not. This raises the question whether the type of nudge matters in situations that are already cognitively demanding. The present study investigated the possible effect of cognitive load on the effectiveness of a type 1 and a type 2 nudge and whether this has an influence on the perceived autonomy. This was done creating a questionnaire about people's travel preferences, in which participants had the option to choose either a sustainable answer option or a less-sustainable answer option. A default nudge was used in the type 1 nudge condition, whereas a positive feedback nudge was used in the type 2 nudge condition to favour the sustainable answer option. A dot memorization task was used to manipulate cognitive load. The results revealed that there was no significant interaction effect of cognitive load and the type of nudge on choosing the sustainable answer options. Furthermore, cognitive load or the type of nudge did not have any influence on the perceived autonomy. These findings hopefully contribute to a positive attitude towards nudging and to the understanding of the underlying working mechanisms as well, so that nudging can be properly applied as a behavioural change tool.

Name company or University: Utrecht University

UU Supervisor: Dr. R.J. Weijers

Name Student: Emmie van Zijtveld

Internship or Thesis: Thesis

Title project: Rhythm perception and its mechanisms: Using neural resonance theory and predictive processing theory to explain human temporal expectations.

Within rhythm perception and cognition there are two theoretical frameworks of temporal expectations. The first framework is Neural Resonance Theory, which argues that we look for periodicity, making beat-based expectations. The second framework is predictive processing theory, which argues that we use statistical regularities, making pattern-based expectations. This study looks into the effect of beat-based expectations, pattern-based expectations and musical expertise on performance of an implicit, explicit and motor task. All tasks were performed on a set of 56 rhythms, divided into four categories: periodic predictable, periodic unpredictable, aperiodic predictable and aperiodic unpredictable. We found that pattern-based expectations seemed accessible for everyone, while beat-based expectations were context dependent. Also, musical expertise increased performance overall, but mostly for more complex rhythms. Unfortunately we cannot make definitive conclusions due to computational difficulties, but it still opens avenues for research into the interplay between beat-based expectations, pattern-based expectations and musical expertise.

Name company or University: Universiteit van Amsterdam

(If applicable) External supervisor: Fleur Bouwer

UU Supervisor: Suzanne Hut

Name Student: Nabila Talitha

Internship or Thesis: Thesis

Title Project: Analysis of Perceived Acceptance and Effectiveness on Default and Social Influenced Nudges Towards Healthier Food Choice: A Cross-Cultural Comparison between Individualist and Collectivist Society

Abstract

In recent years, the impact of climate change on people's lives has increased. It has not only become a problem for large organisations, but over time people have begun to understand that they, as individuals, also have a personal responsibility. As a result, the sustainability movement is increasingly emphasising the concept of personal responsibility, such as people as consumers. One way in which people can not only help the environment but also benefit from it is by switching to more sustainable food choices, such as reducing meat consumption. Behavioural interventions, particularly nudges, have shown promise as tools to encourage this sustainable behaviour. Although research has identified nudging as a promising tool, consumer attitudes toward specific regulatory nudges vary due to individual and cultural differences. The purpose of this study was to analyse whether there is a difference in perceived acceptance and perceived effectiveness between two cultural dimensions, individualistic society and collectivistic society, when given default nudge and a social influenced nudge. Based on the quantitative method used in this study, there is no significant difference in perceived acceptance and perceived effectiveness, however, the value of perceived acceptance is slightly higher for collectivist and social influenced nudge.

Keywords: nudge, culture dimension, acceptance, effectiveness

Name company or University: Utrecht University

Supervisor: Dr. Robert Weijers

Student: Anzhelika Sosnova

Project: Thesis

Title project: Understanding Perception of Algorithmic Predictions

Abstract

Predictive analysis is increasingly done by algorithms including artificial intelligence, deep learning, and other extremely complex models. Perceptions of algorithms' performance can significantly influence their adoption, yet we do not fully understand how people perceive forecasts made by AI as compared with forecasts made by humans. To explore perceptions of algorithmic forecasting, we conducted an online experiment using a set of choice problem with a probabilistic forecast imbedded in each of them. We manipulated the forecast maker (AI or human), and observed preferability of identical options, associated with decision-making under uncertainty cognitive biases, and measured perceived trust in the forecast-maker. In both domains (serious and trivial) the participants demonstrated higher preferability of high confidence forecasts in comparison with medium and low confidence forecasts. We also register preferability of positively framed choice options as opposed to negatively framed choice options. The results are more prominent in Human condition than in AI condition, though the difference wasn't statistically significant. Perceived trust levels were higher towards a human forecaster than towards an AI forecaster (the difference in levels isn't statistically significant). This research reveals people's lay concepts of algorithmic versus human predictions in a context of decision making under uncertainty.

Thesis place: Utrecht University

UU supervisor: Dr. M. J. Mulder

Shayna Coleman

Thesis

Does Learning Make You Tick?: An investigation on the role of reinforcement learning and working memory in learning from past experiences

Utrecht University

Dr. Irene van de Vijver

Reinforcement learning (RL) and working memory (WM) are involved when learning from previous experiences. Both systems operate from different systematic approaches, have differing constraints, and it has been suggested that the two complex systems work proportionately during the learning process. In addition, components such as language and learning strategies such as chunking can help during the learning process and make new information more manageable and improve memory performance during learning. Language can be a powerful tool to generate explicit knowledge. In comparison, learning strategies such as chunking can help to offset the capacity limits of WM. However, an investigation has not been made to understand the workings of RL and WM with respect to skill acquisition and its reliance on language. In the current study, I chose to investigate the interaction between reinforcement learning and working memory to gain further insight into how the systems operate during the learning process to understand the use of language and its role in task learning.

Name Student: Sofia Balali

Thesis

Title project: Does Conspiracy Thinking Influence Trust in the Government? A Dutch Study

Company: TNO, Soesterberg

External supervisor: Dr. Hans Korteling

UU Supervisor: Dr. Suzanne Hut

Belief in conspiracy theories (conspiracy thinking) has been identified as a reducing factor in institutional trust. As research has operationalized trust as three distinct components (*trust, distrust, mistrust*), we examined to what extent conspiracy thinking is correlated to each component and accounts for its variance. Moreover, we investigated to what extent conspiracy theories influence opinions about specific aspects (*competence, benevolence, integrity*) of government performance. Results (N=404) showed that conspiracy thinking was negatively correlated to trust and positively correlated to distrust and mistrust. Furthermore, conspiracy thinking was negatively correlated to all the aspects of government performance. In addition, results showed that conspiracy thinking accounted more than the control variable of skepticism for the variance in distrust (20%) and mistrust (27%), while skepticism accounted more than conspiracy thinking for the variance in performance. Finally, as conspiracy thinking and skepticism combined account for less than 50% of the variance in the trust components and less than 20% of the variance in the government performance, future research should investigate which factors account for the remaining variances.

Name Student: Jingtao (Bill) Zhang

Internship or Thesis: Thesis

Title project: *Enhancing the Proteus Effect through Motor Imagery in Virtual Reality*

Abstract:

The proposed research aims to examine the effectiveness of motor imagery (MI) tasks in enhancing the embodiment level within virtual reality environments, and subsequently, their potential in amplifying the Proteus effect. The study will employ a mixed design, wherein participants will embody either an athletic or obese avatar in two separate VR sessions. The embodiment protocol comprises of forearm movements and head movements, designed to establish a sense of ownership over the virtual body. For one group, a walking-based MI task follows the embodiment, intended to reinforce the sense of embodiment, an embodiment questionnaire is administered for further insights into the effectiveness of the MI task. All participants then complete a food choice task, aiming to measure the influence of the Proteus effect on their decisions regarding food choices. This research endeavors to provide critical insights into the practical application of VR technology in facilitating behavioral change.

Name company or University: Utrecht University

UU Supervisor: Renee Lustenhouwer

Daniel Cotelo García. 8272085. 20/06/2023. Internship.

How modularity level and cover format affect user's willingness to repair their smartphones.

Abstract

It is essential to minimize the negative environmental impact of electronic device manufacturing. This can be achieved by making the product lifetime longer and repairability is a very interesting alternative because only a small number of new pieces are needed, and the device would work as it used to. In this case, we are focusing our research on smartphones. This study pretends to discover how cover format and modularity level affect this new product perception and, therefore, the willingness of the user to repair the product. To test the proposed hypotheses, a 3 x 2 experiment test is conducted. There are three possible levels of the back cover (traditional, transparent, none) and two possible levels of modularity (partial and full). Discussion and conclusion still need to be done to test the hypothesis.

Name of Student: Farah Hamadeh

Internship or Thesis: Thesis

Title project: The Moderating Role of Working Memory Capacity and Signaling on Attention Allocation and Multimedia Learning.

The existing literature on signaling in multimedia learning suggests its potential to enhance learning outcomes. However, the effects of signaling on learners with different working memory capacities (WMC) remain inconclusive. To address this gap, this eye-tracking study examined the moderating effect of WMC on learning outcomes in the presence of signaling cues, while also investigating attention allocation patterns. Fifty participants engaged in a multimedia lesson on synaptic transmission, where signaling cues, including color coding and bolding, were incorporated. Results on the moderating effect of WMC in the presence of cueing did not have a significant effect on learning. However, cueing did demonstrate significant differences in fixation counts and total fixation durations, suggesting attentional differences in response to the cueing manipulation. Moreover, an order effect was observed, underscoring the significance of cue presentation sequence on attention allocation.

Name of company or University: Utrecht University

UU Supervisor: Dr. Margot van Wermeskerken

Name Student: Manfredi Guarisco

Internship or Thesis: Internship (12.5 EC)

Title project: *Optimizing fast-scrolling Advertising in social media: can Content Performance be improved by inducing Attentional Bias? [with the latter addition of:] Profiling exposure to GDPR fines for a better customer journey. Is Digital Maturity related to any of the Big 5 traits?*

Social Media users spend an average of 1.7 seconds to consume one, multimedia content. Here, I tested if following the attentional bias I found in my thesis research (measured in *microsaccades amplitude* and *mnemonic performance*) can improve social media content performance in a fast-consuming case scenario, with A/B testing in LinkedIn Ads. Specifically, I optimized the visual design by implementing changes only in colors and highlighted visual elements positioning, while keeping the same promotional message for both items A and B. I found that content performance for LinkedIn advertisement *did* improve: performance difference for the optimized single-image Ad was widely positive for views (+21.17%) and relatively good for clicks (+8.79%). Performance difference for video Ads was still positive, but thinner. The explorative clients' Digital Maturity modeling and psychological profiling resulted in moderate, positive correlations between Technology, Extraversion and Agreeableness; between Organization, Extraversion and Agreeableness; and in a weak, negative correlation between Technology and Neuroticism. Limitations should be considered for the A/B tests results replicability, for which a pixel-by pixel template has been made to partly counterbalance this issue; and for the Digital Maturity Model and psychological profiling, as for a tool to be considered for qualitative profiling and may still need to follow additional steps for content and/or items validation.

Name company or University: Nixon Digital (Alkmaar)

(If applicable) External supervisor: Ruud Brals

UU Supervisor: Sam van Bijnen

Name Student: Zoe Malvido Montandon

Internship or Thesis: Internship

Title project: Exploring the Influence of Gender, Age, Education, Handedness, and Footedness on the Hand laterality task and the Bergen right-left discrimination Test.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Prof. Dr. Chris Dijkerman

Exploring the Influence of Gender, Age, Education, Handedness, and Footedness on the Hand laterality task and the Bergen right-left discrimination Test.

Recently, the ability to distinguish between left and right has become a highly researched topic. This skill is vital in daily tasks such as driving and can be a crucial requirement in some professions such as for surgeons, with substantial repercussions. The tendency of confusing left and right is common in healthy individuals. As a result, many studies focus on understanding the reasons behind these judgment mistakes. The main goal of this internship was to create an online experiment to assess individual differences in the ability to distinguish between left and right. The main research question was: How do gender, age, level of education, handedness, and footedness affect the ability to distinguish the left hand and the right hand on the hand laterality task and the Bergen left-right discrimination test? The purpose of this project was to develop an online tool to analyze these variables. In addition, this project aimed to contribute to current research on this topic.

Name Student: Mirte Baars

Internship or Thesis: Thesis

Title project: Too costly to care? The influence of the cost of living crisis on sustainable consumer segments

This study introduces a short questionnaire to measure environmentally sustainable lifestyles. In addition, it investigates the impact the cost of living crisis had on people in the different lifestyle segments and explored various drivers and barriers toward a sustainable lifestyle. Previous research has shown that usually women, and higher educated people are more engaged in living sustainably. The new questionnaire was validated by means of a factor analysis and segments were defined by applying a cluster analysis. Three clusters were defined. The clusters did not differ in gender, but there were differences in age, education level, income, and home-ownership. The cost of living crisis has impacted all segments, making them less willing to spend more on sustainable products. The provided questionnaire and drivers and barriers insights can help businesses to more appropriately develop new and existing products and market these to the most fitting group.

Name company or University: Metrixlab

(If applicable) External supervisor: Nele van den Ende

UU Supervisor: Chris Klink

Name Student: Shawn Veldhuis

Internship or Thesis: Thesis

Title project: Examining the Influence of Initial Overtrust on Pedestrian Behaviour in Response to Incongruent Connected and Automated Vehicle Behaviour

Name company or University: German Aerospace Center (DLR)

(If applicable) External supervisor: Marc Wilbrink

UU Supervisor: Dr. Chris Janssen

In this study I investigated pedestrian interaction with connected and automated vehicles that use external human-machine interfaces (eHMI) in mixed traffic environments to enhance pedestrian safety. I assessed pedestrians' interaction with incongruent vehicle behaviour (contradicting eHMI and vehicle kinematics) and the influence of initial trust on willingness to cross in the Netherlands and the UK. I used a mixed 2x2x2x3 within-subject research design with manipulated vehicle size (Car vs Bus, vehicle), Kinematics (Yielding vs non-yielding), and eHMI status (no eHMI vs Static eHMI vs Dynamic eHMI) within participants, with countries serving as a between-subject factor.

The study linked congruent vehicle behaviour with perceived safety and trust in pedestrians. Initial overtrust increased pedestrians' willingness to cross under both congruent and incongruent scenarios, irrespective of vehicle size or country. The eHMI effectively fostered confident crossing, though responses varied between the Netherlands and the UK, underlining the possible difficulty of universal traffic rules (if the effects are not spurious). The study advised future research on trust levels in varied communication scenarios and urged for more realistic experimental settings.

Name Student: Raphael Thorwart

Thesis

Title project: How do emotional engagement and consideration in chatbot behavior impact users' perception of chatbot intent?

Abstract:

This research project aims to investigate the impact of chatbot characteristics on users' perception and trust in human-computer interaction. The experiment will examine the effects of two chatbot conditions: factual and emotionally engaging. The research question focuses on whether users perceive the chatbot as having its own intent. Participants will engage in interactive tasks and a Prisoner's Dilemma game with the chatbot. The study will be conducted online using the Gorilla Experiment Builder platform. The findings will contribute to understanding the role of chatbot characteristics in shaping users' perception and trust, with implications for designing more effective and engaging human-computer interaction systems.

University: Utrecht University

Supervisor: Baptist Liefoghe

Name student: Ydo Baarda (6196543)

Internship or Thesis: Internship

Title project: The effect of palate cleansers on the salt taste perception of expert-tasters

The aim of this internship was to investigate the use of palate cleansers in the research performed by Essensor, a company focused on the improvements of food products. The goal of a good palate cleanser is to establish and maintain a baseline for the taste palate. Research on palate cleansers suggests that the use of specific palate cleansers for certain tastes can be beneficial, reducing the adaptation for said tastes over sequential samples. However, these experiments often report contrasting results. To find a definitive answer for this subject, Essensor made their expert panel with trained supertasters available for experiments. The current project was to establish a procedure for said experiments to be performed over several months. By subjecting the expert panel to several sessions wherein the panelists had to taste gelatin samples containing a certain concentration of salt, a pilot study was conducted.

Name of company: Essensor

External supervisor: Eijnsman, S.

UU Supervisor: Dijksterhuis, G.

Name Student: Carmen van Klaren

Internship or Thesis: Thesis

Title project: "On the right track: Factors determining travel decisions between plane or train on short-distance journeys"

The global air traffic passenger count has steadily risen, contributing significantly to greenhouse gas emissions. Opting for international train travel reduces CO₂ emissions by 80-90%, which is especially feasible for short-distance journeys as it combats a higher fuel consumption per kilometre and the rail network is competitive to air. Despite flight shame and competitive alternatives such as trains, air travel continues to increase, creating a gap between climate concerns and behaviour. This internship at the Royal Netherlands Aerospace Centre (NLR) aimed to identify factors influencing pro-environmental travel choices between trains and planes for short-distance trips from the Netherlands. To achieve this, a questionnaire was developed considering environmental attitude, socio-demographic variables, and travel attributes. A pilot study assessed the questionnaire's effectiveness and generated a dataset comparing individuals traveling to short-distance destinations by air or rail that can serve as a foundation for future research on sustainable travel behaviour at NLR.

Name company or University: Royal Netherlands Aerospace Centre (NLR)

External supervisor: Naomi Sieben

UU Supervisor: Sam van Bijnen

Poster session abstract
Julia Schout
6222897

19-06-2023

Name Student: Julia Schout

Internship or Thesis: Thesis

Title project: Predicting cognitive failures in impulse control

Abstract:

This abstract describes a research project that involved methodological modifications after the individual base regressions produced insignificant results, resulting in the initial model being unfeasible. The new model sought to investigate the associations between the Healthy Lifestyle Beliefs Scale (HLBS), the General Health Questionnaire (GHQ), the Mean Reaction Time Variability (MRTV), and the Stop Signal Reaction Time Means (SSRT).

The improved model revealed a significant and substantial connection between GHQ and HLBS, especially when MRTV was taken into account. Furthermore, MRTV and HLBS were found to have a positive association. The interaction term HLBS:GHQ demonstrated a relationship between psychological discomfort and healthy lifestyle ideas. The resulting model had a good fit and explained a substantial amount of the variability in HLBS.

The importance of including psychological discomfort as a component of healthy lifestyle beliefs is emphasized in this study, as is the potential influence of cognitive processes in forming these relationships. Future studies should look into the moderating effects of self-efficacy and cognitive capacities, as well as longitudinal studies to acquire a better understanding of the dynamic links between psychological discomfort and healthy lifestyle beliefs.

Name company or University: Utrecht University

(If applicable) Second Assessor: Charlotte van de Wouw

UU Supervisor: Daan van Rooij

Name Student: Alisso Andrikopoulou

Internship or Thesis: Thesis

Title project: Artificial prosthetic vision for navigation: finding crucial design parameters with phosphene-based simulation in Virtual Reality

This study aspires to discover indications for optimal design parameters regarding visual implants, with the use of Simulated Prosthetic Vision and the focus on navigation. Visual implants are emerging neurotechnology solutions for the blind that aim to create a rudimentary form of sight by eliciting “phosphenes” (spots of light). Seventeen naïve participants navigated in VR, firstly rating a number of statements regarding user experience with various manipulated design parameters and then 16 of them completed wayfinding trials, measuring their performance across two different phosphene numbers and three levels of brightness. The results show that individuals performed better under the high number of phosphenes, while also preferred it. The level of brightness does not seem to affect their performance. This finding suggests that participants can equally navigate in high and low brightness, leading to the assumption that visual prosthesis devices can be efficient with less current passing through their electrodes, decreasing power consumption and optimizing their total use. As for the user experience brightness is found to differentiate the overall evaluations of the participants.

Name company or University: UU

(If applicable) External supervisor: (not applicable)

UU Supervisor: Chris Klink

Name Student: Surya Tjio

Internship or Thesis: Internship

Title project: "Uncovering the neural responses to climate change images: A proposal for research on affective climate change images and individual levels of pro-environmental behavior"

Consumers have become increasingly aware of their environmental impact. However, the consumption of plastic-packaged products remains relatively unchanged. This discrepancy between what people believe and actually do is referred to as the attitude-behavior gap. Currently, there is ongoing research focused on studying behavior-change interventions to promote more pro-environmental consumer behavior. This internship contributes to this ongoing research by proposing a study that investigates how people's brain activity changes when they view positive, negative, and neutral nature images related to climate change. This knowledge can aid in developing visually-based behavior-change interventions. For this research proposal, an experiment and a behavioral task were developed. The main experiment measured brain activity as stated before. To look at the relationship between people's current level of pro-environmental behavior and their brain activity, the Consumer Pro-Environmental Behavior Task (cPEBT) was developed to measure individual levels of pro-environmental behavior.

Name company or University: Unravel Research

(If applicable) External supervisor: Nikki Leeuwis

UU Supervisor: Birgit Hasenack

Name Student: Michail Olympios (2524155)

Internship or Thesis: Combined

Title project: Measuring blood flow using a webcam and a FLIR camera: Employing remote photoplethysmography to map vascular bed distributions in superficial skin layers across different body parts during demanding tasks

Name Company or University: University of Utrecht

UU Supervisor: Dr. Marnix Naber

Abstract

Objective: Experimentally test the usage of remote photoplethysmography in measuring blood circulation through facial and bodily skin pulsation during rest, mentally and physically demanding tasks

Background: Italian physiologist Angelo Mosso conducted the first neuro-cardiovascular studies with a plethysmograph almost 150 years ago, showing that blood flow pulse increases as a function of mental effort. This discovery faded in time due to the appearance of newly sophisticated methods (e.g.m fMRI) that sketch the cognitive effects of blood circulation and oxygenation across brain regions, with an advanced spatial resolution. A major drawback to these techniques is that they are practically inconvenient, expensive, and not easily accessible to most of the population. That is why in this study, for the first time, we assessed blood flow using remote photoplethysmography (rPPG), a technology that allows contact-free tracking of an individual's cardiovascular changes, frame by frame, from camera footage. Two state-of-the-art algorithms, plane orthogonal-to-skin (POS) and independent component analysis – Naber (ICA) algorithms were performed, as well as a commercial pulse oximeter as ground truth.

Methods: 18 healthy participants were recruited, and face-body recordings were made accordingly. Participants had to perform three mentally demanding tasks, the Stroop task, Arithmetic task and Pitch task as well as two physically demanding tasks, a light and medium-level burpee exercise. Every participant had to stand in front of a black wall looking at the cameras while performing each task independently.

Prospective results: We speculate that more blood will be allocated to the different skeletal muscles during physical exertions, whereas while executing a mental task, a large concentration of blood will be transferred to the brain. We associate an increased skin pulsation at different body parts and a decreased pulsation at the face skin surface in physical exercise, while the inverse effect aspires to be detected during cognitive stress. Moreover, the task-dependent gradient of blood flow and heart rate will be explored through rest, two gradual physical exercises, and three cognitively demanding tasks. A high correlation between rPPG and the ground truth in the measurement of heart rate is expected.

Prospective conclusion: rPPG shows potential indications to capture the blood circulation of the face and body and blood flow during various demanding tasks as well as the heart rate.

Note: The project is in the Analysis phase and results will be extracted in the following weeks